EASY · ALGEBRA

Algebra

13

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 ALGEBRA

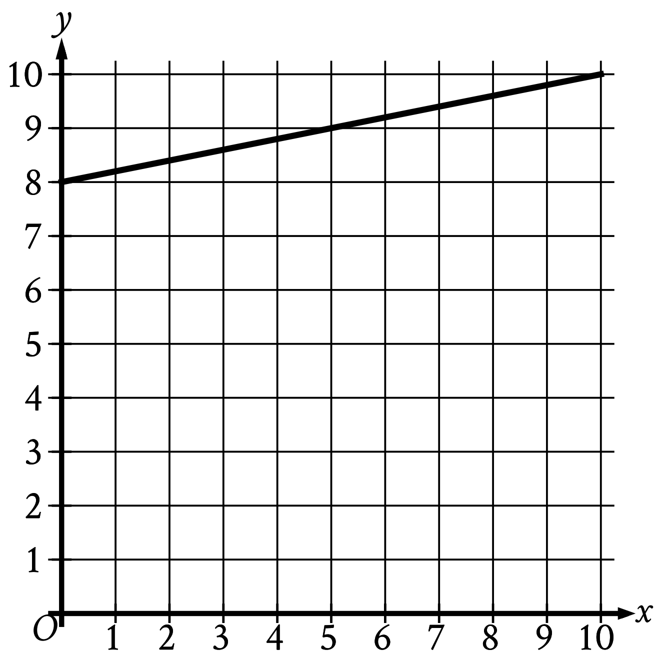
Ty set a goal to walk at least 24 kilometers every day to prepare for a multiday hike. On a certain day, Ty plans to walk at an average speed of 4 kilometers per hour. What is the minimum number of hours Ty must walk on that day to fulfill the daily goal?

A. 4

B. 6

C. 20

D. 24



- The line slants gradually up from left to right.
- The line passes through the following points:
    - (0 comma 8)
    - (5 comma 9)
    - (10 comma 10)

What is the y-intercept of the line graphed?

A. $0, -8$

B. $0, -\frac{1}{8}$

C. $0, 0$

D. $0, 8$

Q3

GRID-IN ALGEBRA

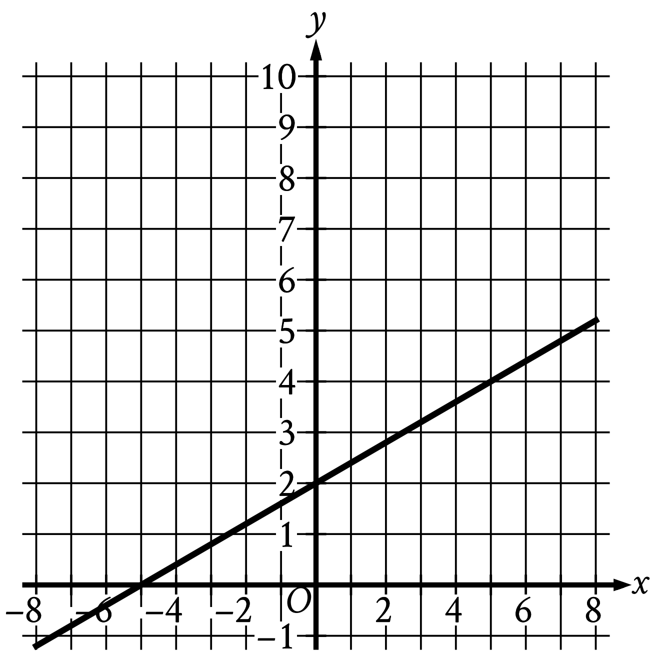
$$13x = 112 - x$$

What value of x is the solution to the given equation?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



- The line slants gradually up from left to right.
- The line passes through the following points:
  - (negative 5 comma 0)
  - (0 comma 2)

The graph of the linear function f is shown. What is the y-intercept of the graph of $y = fx$?

A. -5, 0

B. 2, 0

C. 0, 2

D. 0, - 5

Q5 ALGEBRA

$$x + y = 18$$

$$5y = x$$

What is the solution x, y to the given system of equations?

A. $15, 3$

B. $16, 2$

C. $17, 1$

D. $18, 0$

Q6

ALGEBRA

For a literature class, a student's essay needs to have no more than 5,000 words. If a student's essay already has 3,026 words, what is the maximum number of additional words the student can include in the essay?

A. 1,974

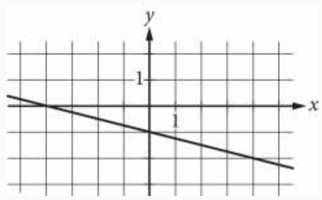
B. 3,026

C. 5,000

D. 8,026

Q7

ALGEBRA



Which of the following is an equation of the graph shown in the xy-plane above?

A. $y = -\frac{1}{4}x - 1$

B. $y = -x - 4$

C. $y = -x - \frac{1}{4}$

D. $y = -4x - 1$

Q8

GRID-IN ALGEBRA

If $2 + x = 60$, what is the value of $16 + 8x$?

STUDENT-PRODUCED RESPONSE

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9

ALGEBRA

Sean rents a tent at a cost of $\$ 11$ per day plus a onetime insurance fee of $\$ 10$. Which equation represents the total cost c, in dollars, to rent the tent with insurance for d days?

A. $c = 11d + 10$

B. $c = 10d + 11$

C. $c = 11d + 10$

D. $c = 10d + 11$

Q10 ALGEBRA

$$4x - 3y = 5$$

$$x = 8$$

What is the solution x, y to the given system of equations?

A. $8, 9$

B. $8, -24$

C. $8, -9$

D. $8, 24$